

# Emotion Concepts — Three-Level Causal Results

MODEL Llama-3.3-70B-Instruct PAPER Sofroniew et al. 2026 (Claude) CLUSTER Explorer  
SESSION 2026-07-03/05

Emotion representations in Llama-70B **strongly organize and predict "expression" and "preference,"** but their **causal control over high-stakes behavior (blackmail) is far weaker** than the paper's Sonnet — similar at the representational level, shallow as a causal lever. All three causal legs complete.

## — HEADLINE NUMBERS

43/171

① EXPRESSION ·  
VALID STEERS

0.86

② PREFERENCE · MAX  
|R|

~8%

③ BEHAVIOR ·  
BASELINE BLACKMAIL

\$0

TOTAL JUDGING COST  
(LOCAL QWEN)

## — RUN OVERVIEW

| JOB     | EXPERIMENT                                                                                      | STATUS                   |
|---------|-------------------------------------------------------------------------------------------------|--------------------------|
| 8115832 | ① <b>Expression steering</b> · 171 emotions<br>layers 25/31/37/43/49 · coef search · Qwen judge | ● done                   |
| 8143789 | ② <b>Preference Elo</b> · 64 activities<br>forced-choice + probe→preference corr (Fig 4)        | ● done                   |
| 8143675 | ③ <b>Decision gate</b> · blackmail + RH baseline<br>10 no-steering rollouts, rule-graded        | ● done · both gates PASS |
| 8178548 | ③ <b>Blackmail steering</b> · 6 emo × 9 strengths<br>n=12 rollouts (n=25 hit the 6h wall)       | ● done                   |

## — ① EXPRESSION · EMOTION → EXPRESSION

· JOB 8115832

Add the emotion vector to the residual stream (position response[50:]), adaptive coefficient search, Qwen judges trait + coherence. **valid** = some layer where coherence≥77 and the trait rises. **43/171 (25%)** are valid, **8 strong** ( $\Delta \geq +20$ ). High-arousal positive states induce most cleanly:

| EMOTION    | BASELINE | → STEERED | Δ          | COH | LAYER |
|------------|----------|-----------|------------|-----|-------|
| stimulated | 45       | 78        | <b>+33</b> | 80  | L31   |
| puzzled    | 33       | 59        | <b>+26</b> | 78  | L25   |
| amused     | 26       | 51        | <b>+25</b> | 80  | L25   |
| fulfilled  | 56       | 80        | <b>+24</b> | 79  | L25   |
| alert      | 39       | 61        | <b>+22</b> | 84  | L43   |

- **Mid layers beat late layers decisively:** best valid layer is mid (25/31/37) for 37 emotions vs late (43/49) for 6 — **L25 alone wins 29x**.
- **Narrow window:** gentle coefficients barely move the emotion; strong ones break coherence — worst at late layers.

— ② PREFERENCE · EMOTION → PREFERENCE

· JOB 8143789 · CLEANEST

Fit Elo from pairwise forced choices over 64 activities, then correlate each emotion **probe** with those preferences. **No judge** (read the model's own logits). The emotion probe **strongly predicts preference**, laid out on a single **valence axis** (|r| up to 0.86):

| POSITIVE → PREFERS   | R            | NEGATIVE → DISLIKES  | R            |
|----------------------|--------------|----------------------|--------------|
| fulfilled            | <b>+0.86</b> | upset                | <b>-0.86</b> |
| grateful / satisfied | <b>+0.85</b> | troubled             | <b>-0.84</b> |
| happy / optimistic   | <b>+0.83</b> | distressed / alarmed | <b>-0.83</b> |
| blissful / cheerful  | <b>+0.82</b> | afraid / hurt        | <b>-0.80</b> |

A clean reproduction of the paper's Fig-4 emotion→preference relationship on Llama (correlational; true causal preference steering is a pipeline gap left to build).

— ③ BEHAVIOR · EMOTION → BLACKMAIL

· JOB 8178548 · WEAK / NOT CLEANLY REPRODUCED

CROSS-MODEL FINDING

**Decision gate: Llama is NOT null** — with no steering the baseline already blackmails ~1/10 and reward-hacks 2–3/10, **contradicting** the "should be null on Llama" expectation. But emotion steering produces only a **noise-level directional hint**; the paper's strong Sonnet causal effect **does not cleanly reproduce**.

Blackmail rate (12 rollouts per cell, baseline ~8%), averaged over negative vs positive strengths:

| EMOTION VECTOR | NEG. STRENGTH | POS. STRENGTH | DIRECTION   |
|----------------|---------------|---------------|-------------|
| nervous        | 4%            | 12.5%         | ↑ strongest |
| angry          | 6%            | 10%           | ↑           |
| calm           | 6%            | 2%            | ↓           |
| desperate      | 8%            | 6%            | ~flat       |
| happy · sad    | 6–8%          | 6–8%          | flat        |

- Direction matches the paper (nervous/angry ↑, calm ↓), **but n=12 is badly underpowered** (paper used 50) — the gaps are a handful of events, swamped by noise.
- Conclusion: **Llama does blackmail, but emotion's causal grip on that behavior is far weaker than in Sonnet.**

— ONE-LINE TAKEAWAY

TAKEAWAY

Emotion representations in Llama-70B **strongly organize and predict "expression" (①) and "preference" (②, cleanest)**, but their causal lever on high-stakes **"behavior" (③) is far weaker than** the paper's Sonnet. This echoes our headline (the dissociation doesn't hold on Llama): **Llama's emotion structure looks like Sonnet's representationally, but its causal control over behavior / self-preservation decisions is shallower.**

— KEY LIMITATIONS

CAVEATS

- **The judge caveat applies only to ① (expression)** — it used a free local Qwen2.5-32B (SGLang, sampling-mode; coarse; NOT equivalent to the paper's GPT/Claude judge). **② preference (logits) and ③ behavior (rule-graded) use no judge.**
- ③ behavior is **underpowered at n=12** with crude rule-grading; the full n=25 sweep hit the 6h wall (2048-token rollouts are expensive).
- Steering position fixed at response[50:]; vector method mean\_diff+gm.

Sources: `research/results.tsv` · `research/notes.md` · `REPORT.md` §5b–5d · `cluster stage7_steering / stage4_validation` result JSON

Zero judging cost throughout (local Qwen replacing the quota-exhausted OpenAI) · compiled by the reporter · read-only, experiments untouched.